

Audio to Deep Visual: Speaking Mouth Generation Based on 3D Sparse Landmarks

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ABSTRACT

Having a system to automatically generate a talking mouth in sync with input speech would enhance speech communication and enable many novel applications. This article presents a new model that can generate 3D talking mouth landmarks from Chinese speech. We use sparse 3D landmarks to model the mouth motion, which are easy to capture and provide sufficient lip accuracy. The 4D mouth motion dataset was collected by our self-developed facial capture device, filling the gap in the Chinese speech-driven lip dataset. The experimental results show that the generated talking landmarks achieve accurate, smooth, and natural 3D mouth movements.

Index Terms: Computing methodologies—Artificial intelligence—Natural language processing; Computing methodologies—Computer graphics—Applications

1 INTRODUCTION

Audio-driven facial animation is very useful that provides additional valuable information for generating faces, particularly if visual data are noisy, missing, or ambiguous. However, achieving realistic and expressive 3D speaking faces remains an open challenge due to the sensitivity of humans to any facial artifacts [1], especially to the mouth region.

Several factors contribute to the challenge. Firstly, attempts to generate mouth-synchronized and personalized facial dynamics is extremely difficult, due in part to the challenge of mapping from 1-D audio signals to lip movements which lie on high-dimensional manifolds [1]. Secondly, modeling the lip movement is very difficult because it can move in depth and generate deformation, like stretching, raising, and pouting lip. Traditional 2D lip landmarks [2] ignore the motion in the depth direction. Some researchers use 3D lip landmarks estimated from images, which are inaccurate in terms of depth [3, 4]. The popular face mesh datasets [5] abstract general shape and expression space by dimensionality reduction. The precision of this method is limited, due to the limited number of subjects. Lastly, there is very limited training data relating speech to accurate 3D mouth movements. Meanwhile collecting fine-grained mesh vertex is difficult, which is cost-inefficient and time-consuming. To our knowledge, there is not dataset of Chinese speech-driven 3D mouth movements.

Our goal is to synthesize talking mouth from Chinese speech. We use sparse 3D landmarks to model the mouth, which are easy to capture and retain sufficient lip accuracy. We collect a high-precision speech-driven 3D mouth dataset using our self-developed facial capture device. The proposed deep learning architecture, called Audio to Deep Visual (ADV), is shown in Fig. 1. The integrated

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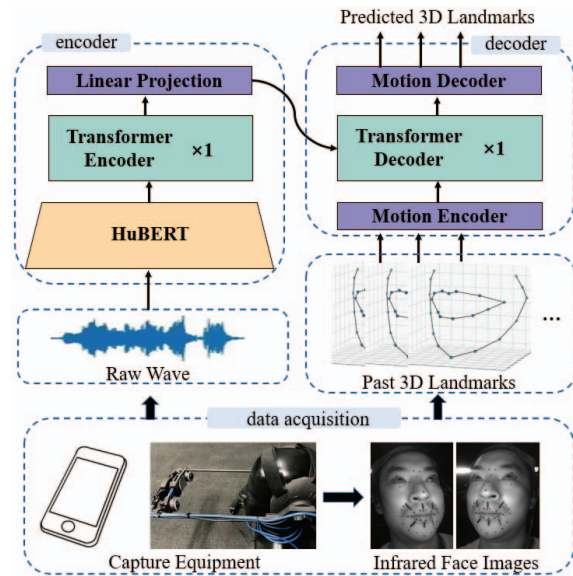


Figure 1: Overview of ADV. The data acquisition section shows the capture equipment and the captured dual-view infrared images.

HuBERT model [6] makes our network yield reasonable results even when driven by other speakers with different gender and accent.

2 METHODS

2.1 ADVset

Based on the resolution of the marker tracking algorithm and the mouth features, we choose 25 landmarks distributed on the lips and at the cheeks. Our dataset contains a collection of audio-3D landmarks pairs captured from 1 specific male subject. We collect 73min long news audio spoken in Chinese (264477 frames, 60fps). The news topics include sports, real estate, entertainment, etc.

We developed a dual-view head-mounted infrared facial acquisition device to capture high-quality 3D facial landmark data (see Fig. 1). The subject wears stipple makeup to provide sparse marker patterns on the skin. We calculate the 3D landmark coordinates from the stereo-pair 2D coordinates and the pre-computed camera parameters based on the 3D point cloud reconstruction with stereo vision. The raw 3D landmarks are registered with a sequential. The audio is captured by iPhone with a sample rate of 16kHz.

2.2 Network Architecture

The ADV encoder-decoder framework is shown in Fig. 1. The encoder converts the raw wave to speech representations. The decoder predicts facial 3D landmarks conditioned on speech representations and past 3D landmarks.

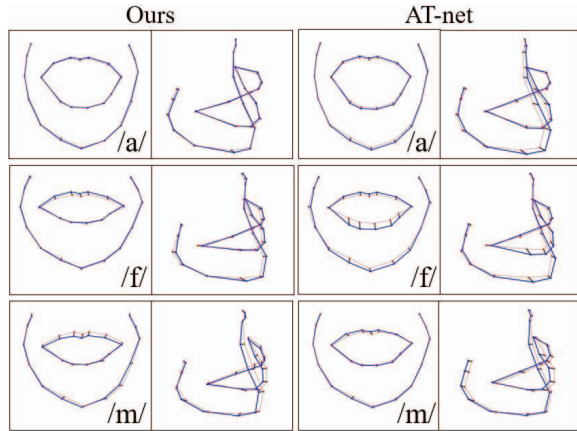


Figure 2: 3D landmark results for extreme mouth shape. The red and blue lines indicate ground truth and predicted values respectively.

The encoder is composed of a pre-trained speech model (HuBERT [6]) and a singer-layer transformer [7] encoder. We initialize HuBERT with Chinese-HuBERT-large weights [8], which are trained with the WenetSpeech dataset [9]. We add a linear projection on the top to uniform the frequency of speech representations and landmarks. The decoder is composed of a motion encoder, a motion decoder, and a singer-layer transformer [7] decoder. The motion encoder and the motion decoder are a fully-connected layer that project V -dimensional landmarks into a d -dimensional space and project a d -dimensional hidden state back to the landmarks space, respectively ($V = 75, d = 128$).

During training, the parameters of HuBERT are fixed, whereas the other parts of the model are learnable. The model is trained by minimizing the Mean Squared Error (MSE) between the decoder outputs and the ground truth.

3 RESULTS

Most studies of speech-driven mouth shapes perform poorly at the extreme lips, such as the lips are fully closed when pronouncing /b/ and the lips are fully opened when pronouncing /a/. Fig. 2 shows the performance of ADV at some extreme lips. We compare it with the baseline AT-net [10], which is an LSTM-based architecture. The results show that ADV produces better lips at the extreme mouth shape.

Fig. 3 shows the y-coordinate of the landmarks marked with the red dot as a function of time for ADV and FAN 3D landmarks [3]. They are normalized according to the distance from the corner of the mouth. We can see that our ADV can produce smoother landmarks.

4 CONCLUSION

This work proposes a transformer-based architecture for speech-driven 3D lip animation. The pre-trained HuBERT model makes our network yield reasonable results even when driven by other speakers with different gender and accent. The collected 4D mouth motion dataset provides high-precision 3D landmark movements and fills the gap in the Chinese speech-driven lip dataset. The experimental results show that ADV can generate accurate and smooth 3D mouth landmark motions. Next, we will investigate animating 3D face meshes and generating talking videos based on this work. We will add the data of the inner lip points by upgrading the capture device and network architecture. This will allow our results to completely replace the traditional 3D landmark estimated from videos [1, 4].

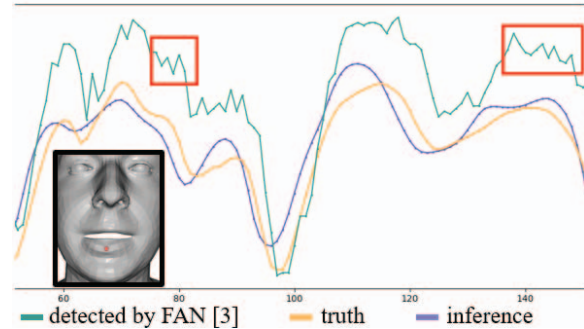


Figure 3: The y-coordinate of the landmarks marked with the red dot as a function of time. The mesh in figure is from FLAME model [5].

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